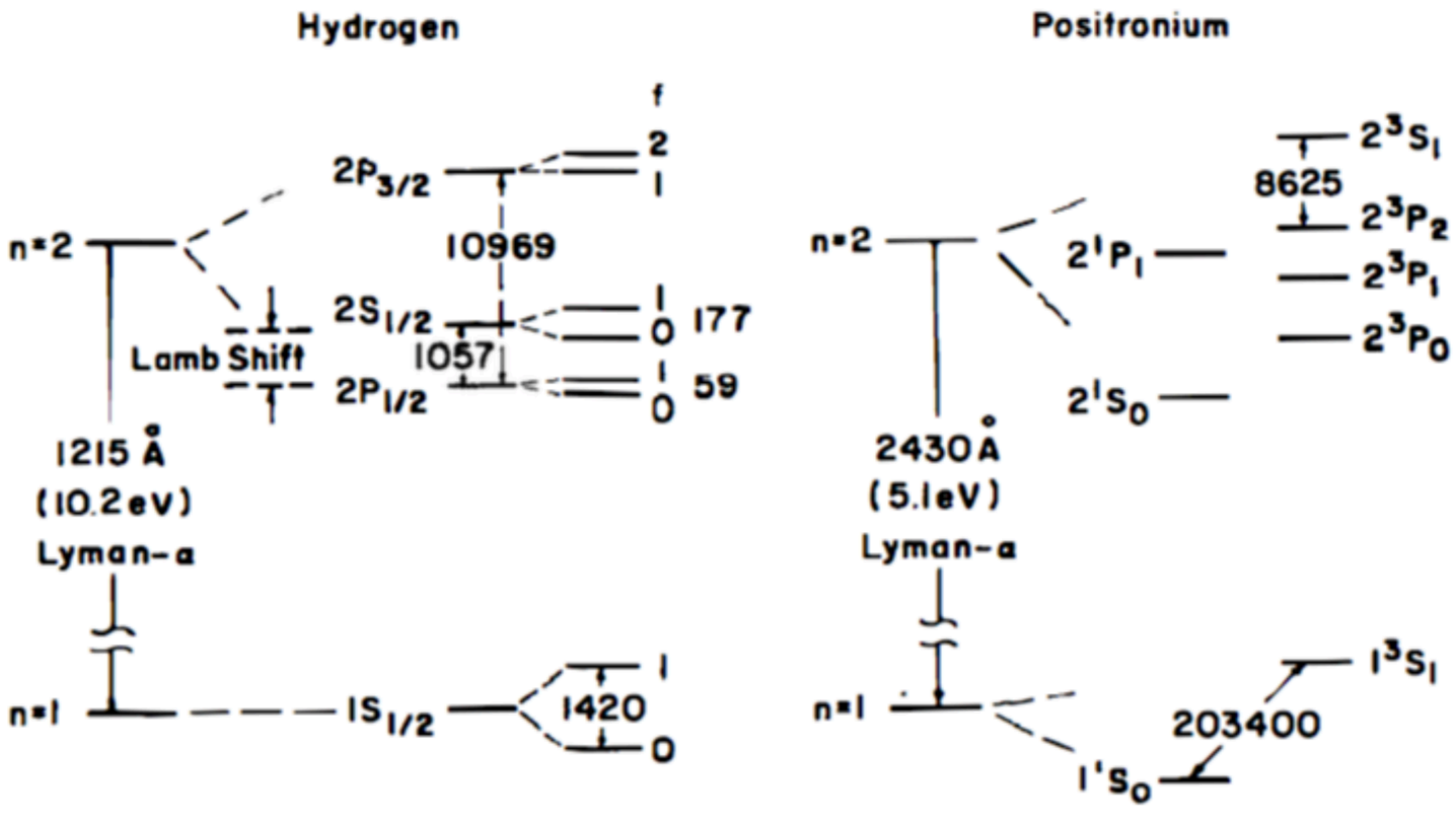


Fig. 4.1: Comparison of the 1S, 2S, and 2P energy levels of hydrogen and positronium, from (Berko and Pendleton 1980).

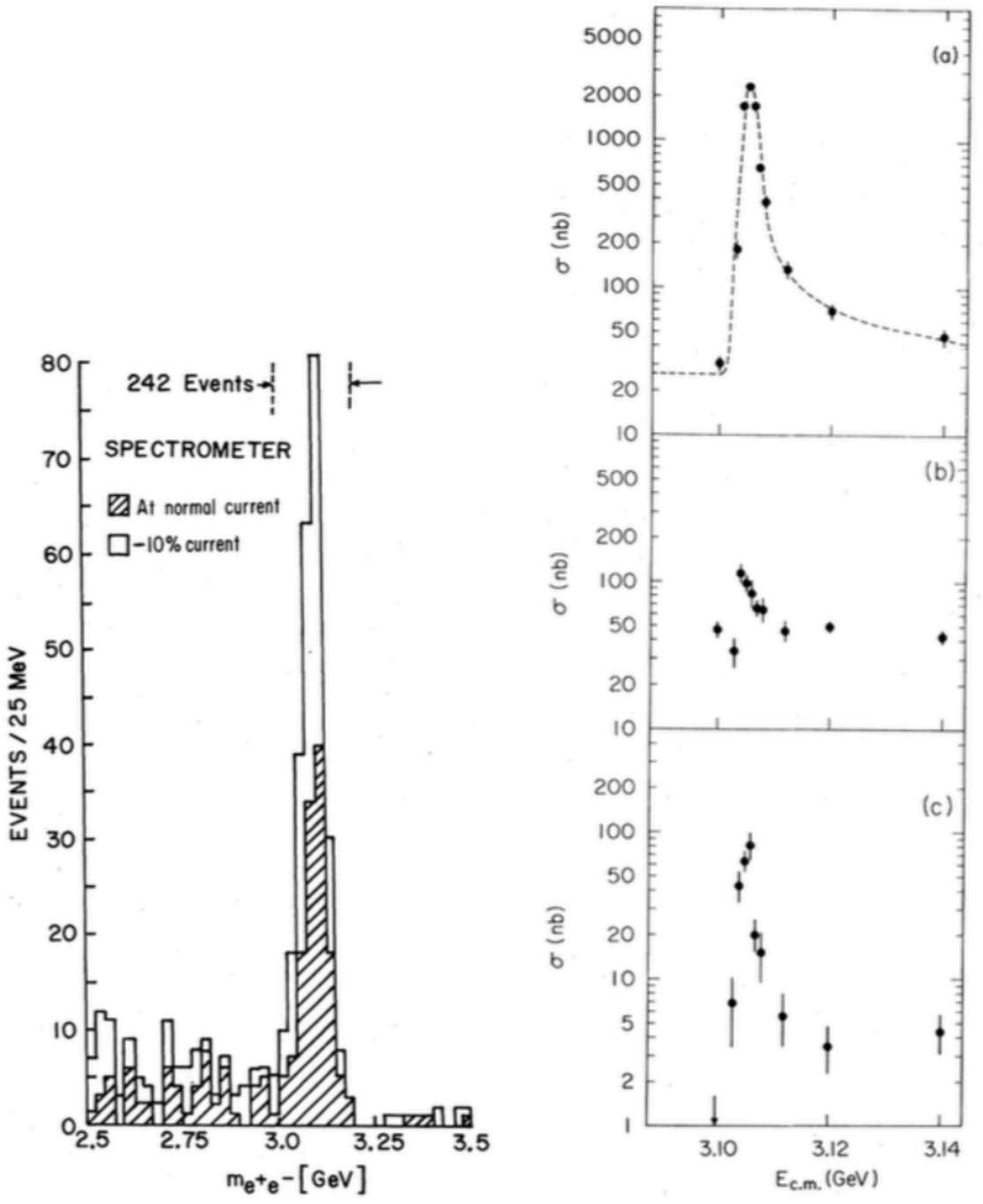


Fig. 5.4: Left: Accumulation of events from the reactions $pp \to e^+e^- + X$ at a fixed value of the e^+e^- invariant mass, proving the existence of the narrow resonance $J\psi$, from (Aubert *et al.* 1974). Right: Observation of a resonance in e^+e^- annihilation near 3.1 GeV, proving the existence of the J/ψ, from (Augustin *et al.* 1974). From top to bottom, the three plots show the production rate of final states with (a) multiple hadrons, (b) e^+e^-, and (c) pairs of μ, π, K.

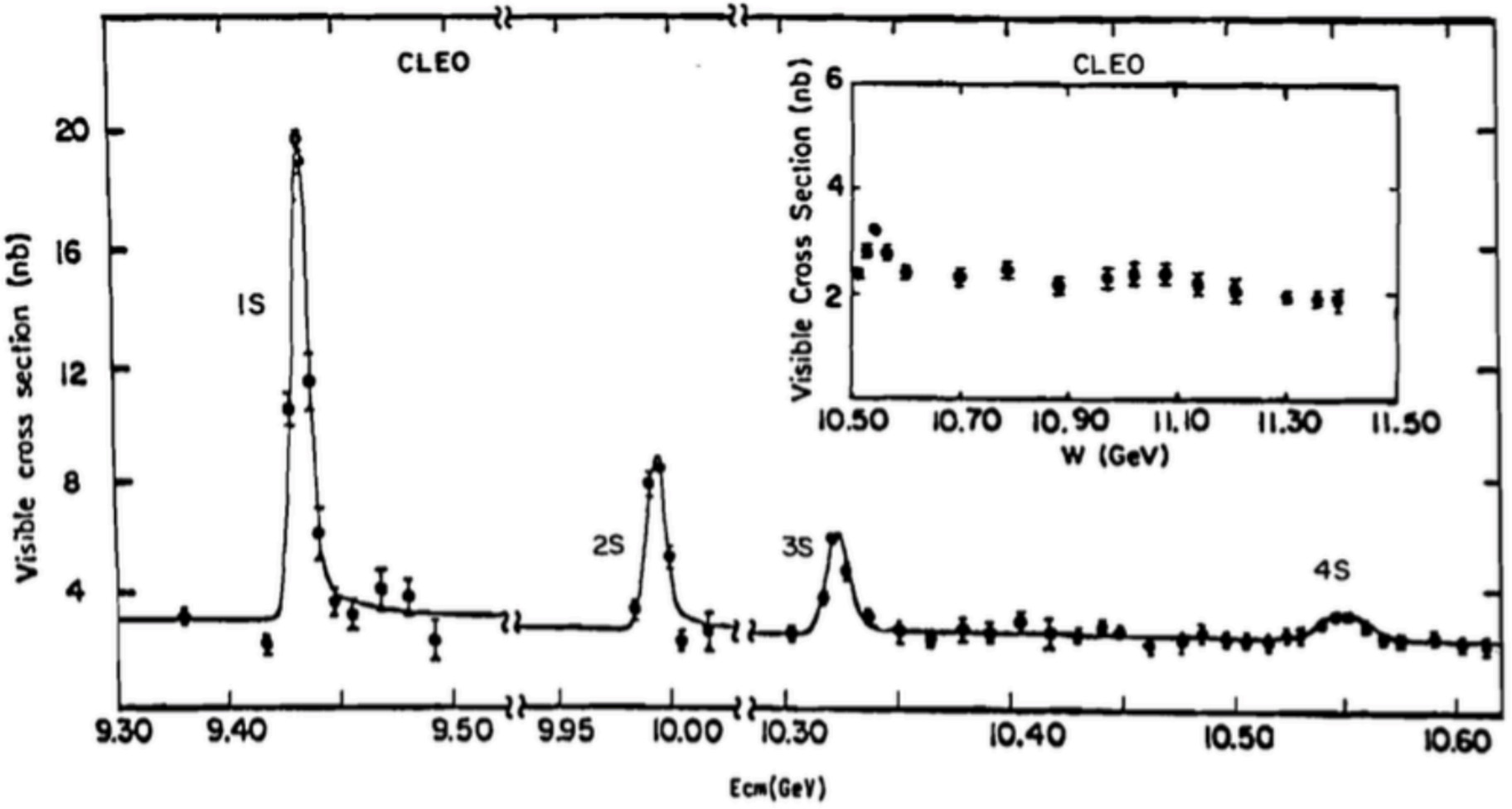


Fig. 5.5: Resonances in e^+e^- annihilation to hadrons corresponding to the S states of the Υ family, from (Silverman 1981). The inset extends the dataset to higher center of mass energy.

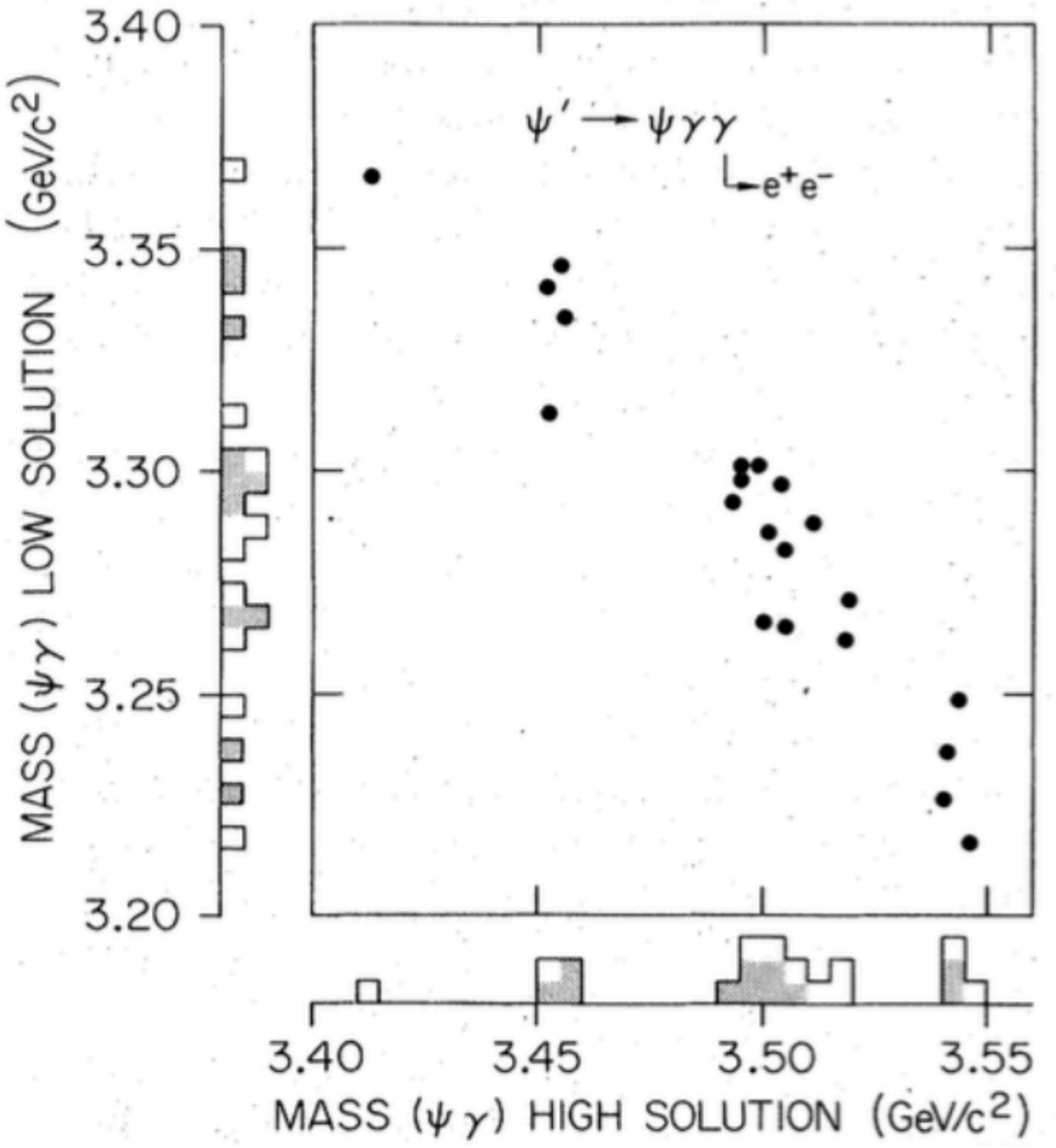


Fig. 5.6: Distribution of $\psi\gamma$ invariant masses observed in the reaction $\psi' \to \gamma\gamma\, J/\psi$ by the Mark I experiment at SPEAR, from (Tannenbaum *et al.* 1978).

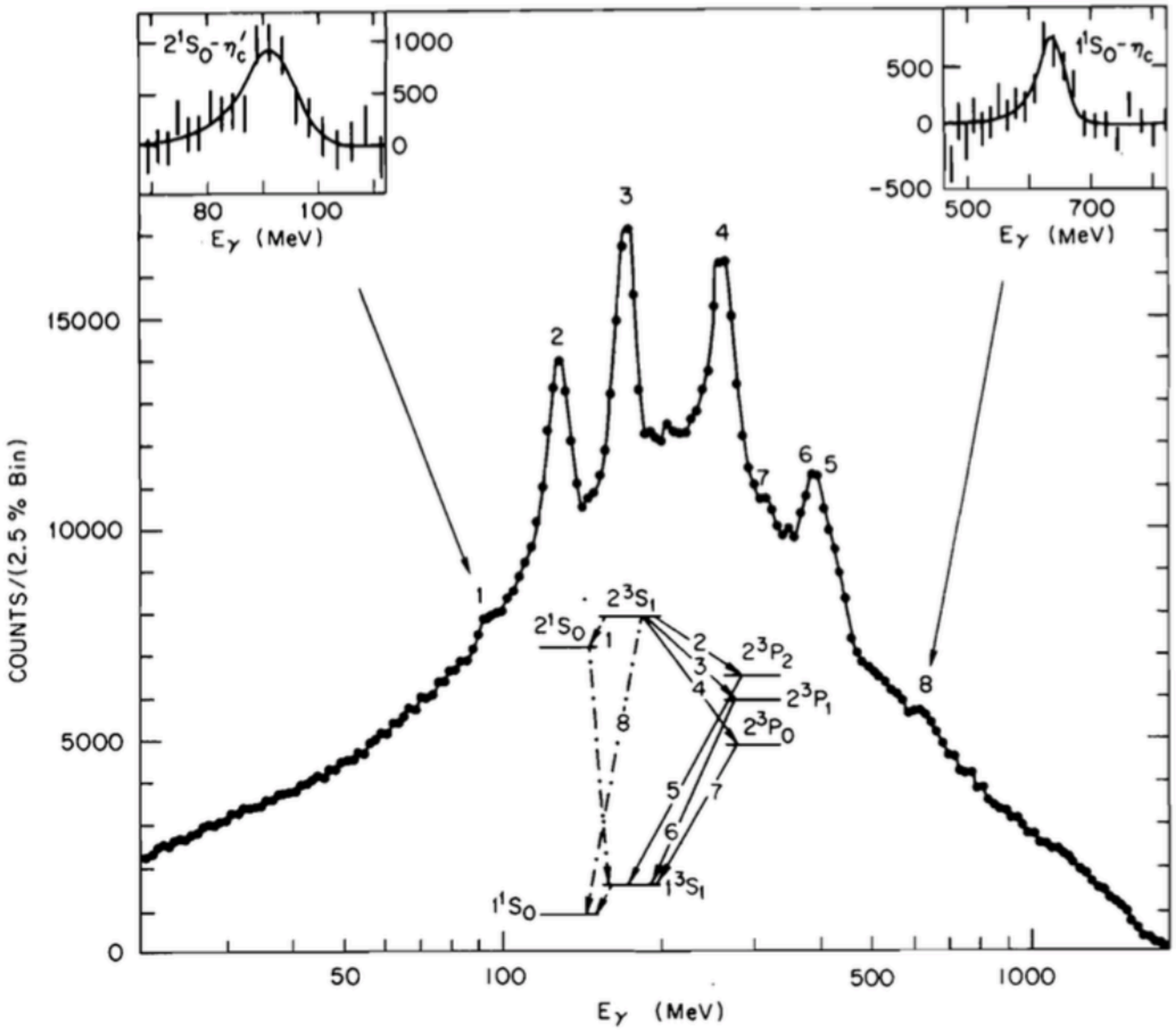


Fig. 5.7: Energy spectrum of photons observed at the ψ' resonance by the Crystal Ball experiment, from (Scharre 1981).

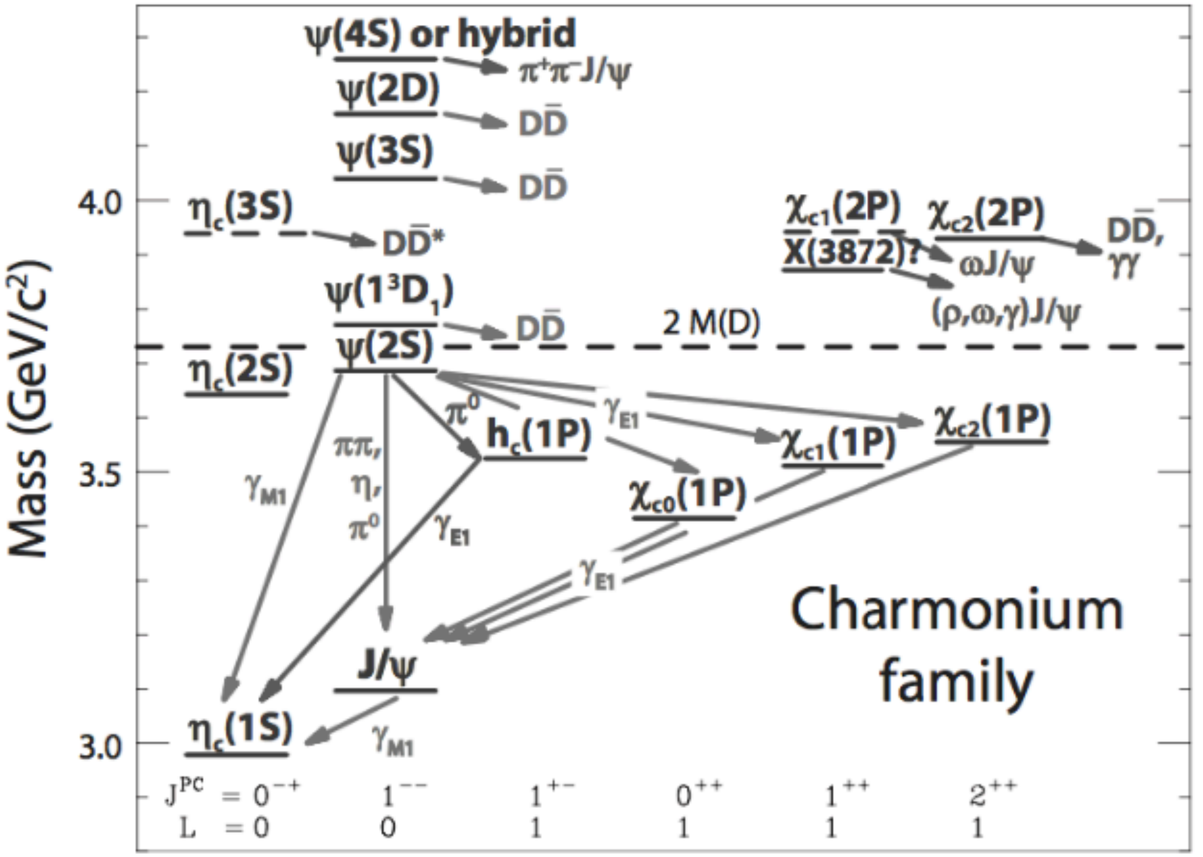


Fig. 5.8: Observed states and transitions of the J/ψ system, from (Eichten, Godfrey, Mahlke, and Rosner 2008). Note that the lowest P and D states are labelled 1P and 1D in this figure, whereas in the hydrogen atom they would be called 2P and 3D. The dashed horizontal line marks the threshold for pair production of D mesons, described in Section 5.4.

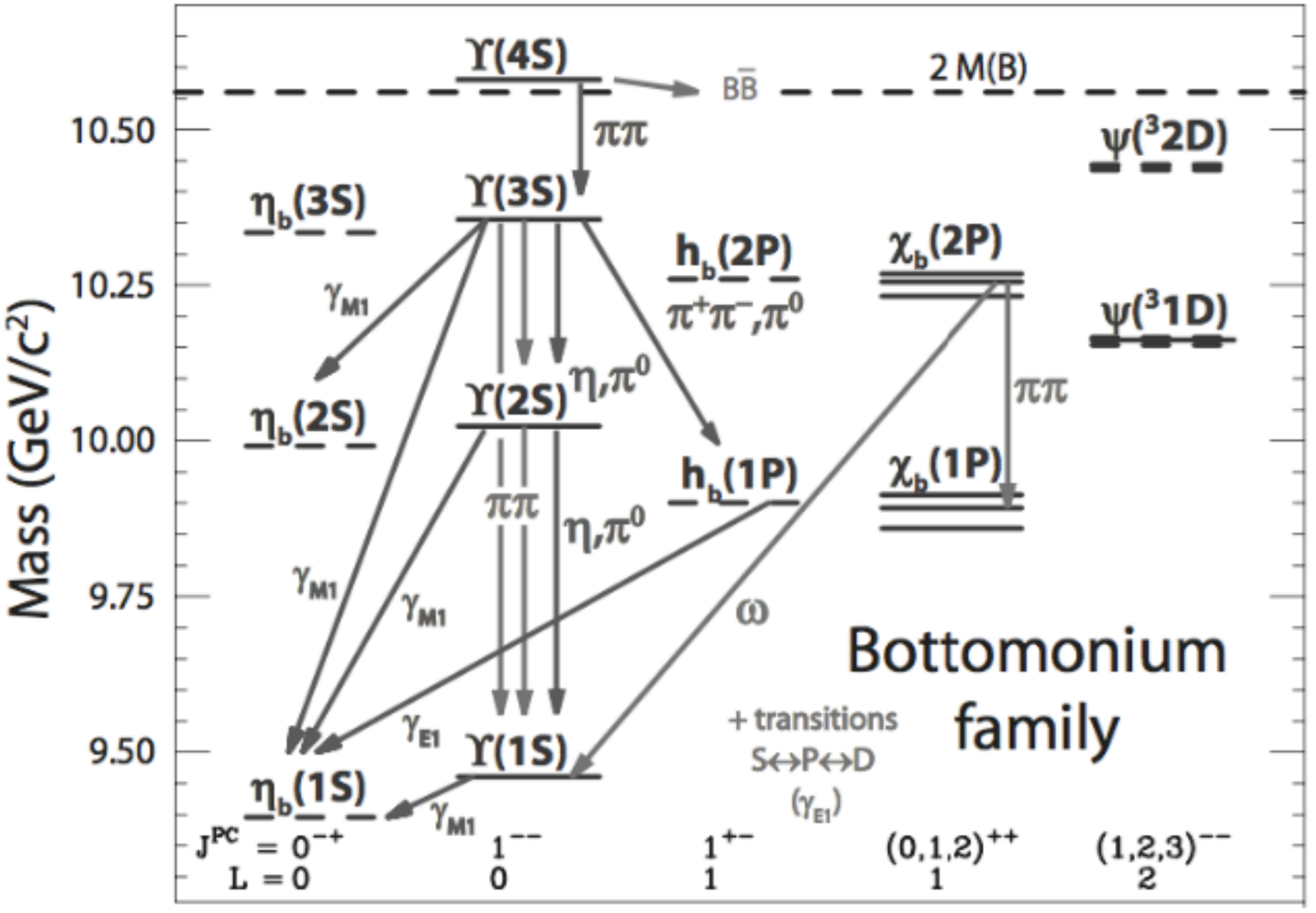


Fig. 5.9: Observed states and transitions of the Υ system, from (Eichten, Godfrey, Mahlke, and Rosner 2008). The dashed horizontal line marks the threshold for pair production of B mesons, described in Section 5.4.